

Experiential Learning Project
Assessment-I on
“POLYNOMIAL ADDITION USING
LINKED LIST”
under the course

Python Programming Language(N-PCCCM304P)

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Problem statement

- Develop a Python program to represent two polynomials using linked lists.
- Each node stores the coefficient, power, and link to the next term.
- Compare the powers of corresponding terms and combine terms with the same power.
- Generate and display the resultant polynomial after addition.
- The project demonstrates the practical use of linked lists and dynamic data structures.

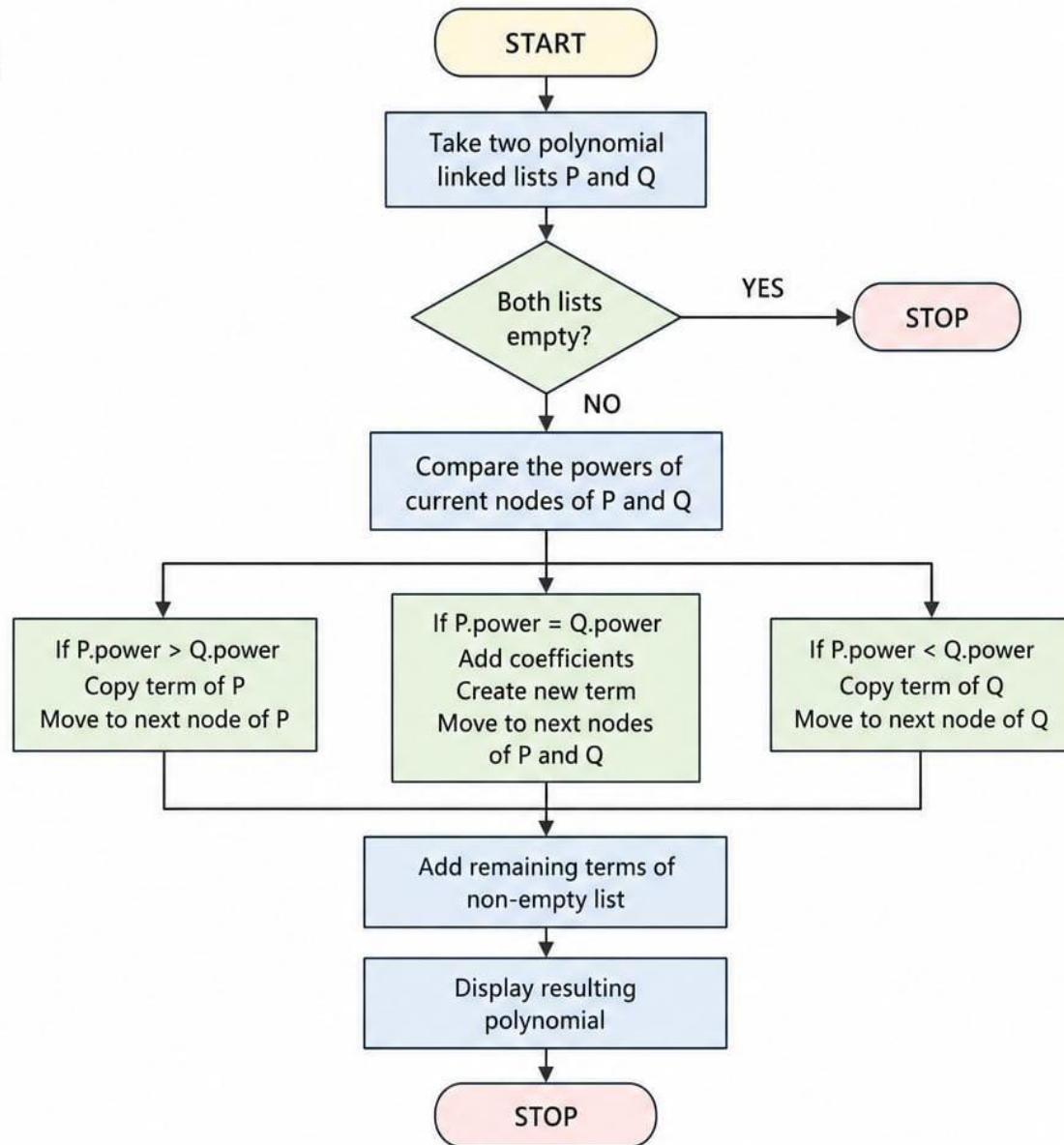
Objectives

- Represent polynomials using linked lists.
- Store the coefficient and power of each term.
- Compare the powers of two polynomial terms.
- Add coefficients of terms having the same power.
- Generate and display the resultant polynomial.
- Understand the use of linked-list traversal in Python.

Algorithm

1. Start
2. Take two polynomial linked lists P and Q.
3. Compare the powers of the current nodes.
4. If powers are equal, add their coefficients.
5. If $P.\text{power} > Q.\text{power}$, copy the term of P.
6. If $P.\text{power} < Q.\text{power}$, copy the term of Q.
7. Move to the next node and repeat the process.
8. Add the remaining terms of the non-empty list.
9. Display the resultant polynomial.
10. Stop.

Flowchart



Sample Input and Output

Input

Enter Polynomial 1
Enter number of terms: 3
Enter coefficient: 5
Enter power: 3
Enter coefficient: 4
Enter power: 2
Enter coefficient: 2
Enter power: 0

Enter Polynomial 2
Enter number of terms: 3
Enter coefficient: 3
Enter power: 3
Enter coefficient: 2
Enter power: 1
Enter coefficient: 7
Enter power: 0

Output

Polynomial 1:

$$5x^3 + 4x^2 + 2x^0$$

Polynomial 2:

$$3x^3 + 2x^1 + 7x^0$$

Resultant Polynomial:

$$8x^3 + 4x^2 + 2x^1 + 9x^0$$

Understanding of Python Logic

- Node class stores the coefficient, power, and next node.
- Two linked lists P and Q represent the input polynomials.
- The program compares the powers of the current nodes.
- If powers are equal, their coefficients are added.
- If powers are different, the term with the higher power is added first.
- The process continues until both lists are completed.
- The final linked list represents the resultant polynomial.

GitHub Account, Repository Creation & Uploading Code

- Created a GitHub account.
- Created a repository named Polynomial-Addition-Linked-Lists.
- Uploaded the Python file polynomial_addition.py.
- Added a README file containing project information.
- Used meaningful commit messages while uploading the code.
- Tested and verified the uploaded Python program.
- GitHub Repository Link:

https://github.com/nikitabml25-del/Polynomial_Addition-Linked-Lists.git

Conclusion

- Polynomial addition can be efficiently implemented using linked lists.
- Each node stores the coefficient and power of a polynomial term.
- The program compares powers and adds coefficients of similar terms.
- Python makes the implementation simple using classes and linked-list operations.
- This project helps in understanding linked lists, traversal, and dynamic data structures

Thank You